Jianning Li

Academic Teaching & Student Mentorship Portfolio

Lecturer at **Free University of Berlin** & **Charité – Universitätsmedizin Berlin**
Researcher at **Zuse Institute Berlin (ZIB)**

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Updated: 2026

01 TEACHING PHILOSOPHY & METHODOLOGY

My instructional approach focuses on building rigorous methodological foundations in machine learning and data science, complemented by practical applications in digital medicine and healthcare. Teaching across computer science and clinical faculties at Free University of Berlin and Charité, I instruct students with diverse backgrounds ranging from computer science to clinical medicine. Coursework and seminars emphasize algorithm fundamentals, reproducible code implementation, statistical validation, and translational deployment of AI for clinical decision support and 3D medical imaging.

Prospective Student Supervision: I regularly supervise Bachelor's and Master's theses in 3D medical shape reconstruction, generative modeling, Bayesian neural networks, and clinical decision support.

02 UPCOMING COURSES (2026)

From Large Language Models to Agentic AI

October – November 2026

Charité – Universitätsmedizin Berlin · Advanced Seminar

An advanced seminar designed for medical researchers, clinicians, and graduate students exploring state-of-the-art developments in generative AI — from foundational prompt engineering with modern LLMs to autonomous agentic workflows and tool-calling architectures in clinical diagnostic pipelines.

Python in the Agentic AI Era: Coding and Prototyping for Non-Programmer Professionals

December 2026

Technical University of Berlin (TU Berlin) · Practical Hands-On Course

A hands-on, practical course tailored for researchers, clinicians, and professionals to harness modern Python alongside agentic AI coding assistants for rapid prototyping, workflow automation, and computational data analysis without requiring prior software engineering backgrounds.

03 UNIVERSITY LECTURES & CORE CURRICULUM

Data Science for Life Science

Co-instructor · Summer Semester 2025

Free University of Berlin · Department of Mathematics and Computer Science

Lectures introducing advanced data analysis pipelines, statistical modeling, exploratory data techniques, and data science methodologies tailored specifically for bioinformatics, biology, and computational life sciences curriculum.

Machine Learning

Co-instructor · Winter Semester 2025

Free University of Berlin · Department of Mathematics and Computer Science

Core curriculum lectures detailing classical and modern machine learning foundations, including optimization algorithms, statistical learning theory, deep neural network architectures, representation learning, and rigorous validation practices.

Statistik für Informatik (Statistics for Computer Science)

Tutor · Summer Semester 2025

Free University of Berlin · Department of Mathematics and Computer Science

Leading exercise classes, computational tutorials, and problem-solving sessions in probability theory, statistical hypothesis testing, distributions, and exploratory data analysis for computer science majors.

04 ADVANCED SEMINARS & SPECIALIZED CLINICAL WORKSHOPS

Creating & Hosting Your Own Dental AI Chatbot
March 2026

Charité – Universitätsmedizin Berlin · 2-part intensive seminar (6 hours)

Guiding dental professionals, researchers, and clinical students through conceptual architecture, API integration, Retrieval-Augmented Generation (RAG), domain knowledge indexing, and cloud hosting for a clinical chatbot.

Artificial Intelligence in Dental Science
Jun 2025 – Aug 2025 (Weekly)

Charité – Universitätsmedizin Berlin · Weekly Seminar Series

Systematic seminar series analyzing state-of-the-art applications of AI, deep learning, and computer vision in dental science, including tooth segmentation, radiograph classification, and diagnostics.

Python Essentials for Medical Professionals: A Hands-On Introduction
February 2025

Charité – Universitätsmedizin Berlin · Two 3-hour seminars (21 – 28.02.2025)

Introductory practical seminar designed specifically for clinicians and medical researchers with zero prior programming experience, focusing on pandas, NumPy, automated data wrangling, and scientific visualization.

Towards Automatization of Cranial Implant Design: Retrospect, Methods & Perspectives
December 2024

Charité – Universitätsmedizin Berlin · ZMK Doctoral Colloquium

Doctoral presentation and discussion covering automatic skull defect reconstruction, Bayesian neural networks for uncertainty quantification, and clinical implant modeling strategies.

05 STUDENT SUPERVISION (THESES & RESEARCH PROJECTS)

Full Spine Reconstruction from Partial 3D Observations
Yunlong Zhao · Master's Thesis (2023 – 2024)
University of Duisburg-Essen

Improving Generalization & Privacy for Cranial Implant Design via Federated Learning
Gabriel Weißenback · Master's Thesis (2022 – 2023)
Graz University of Technology (TU Graz)

Volumetric Shape Super-resolution
Matthias Eder · Master's Project (2021 – 2022)
Graz University of Technology (TU Graz)

Semiautomatic Cranial Implant Design Tool (Rigid ICP & Voxel Reconstruction)
Michael Lackner · Bachelor's Thesis (2022 – 2023)
Graz University of Technology (TU Graz)

Sparse Convolutional Neural Network for Skull Reconstruction
Artem Kroviakov · Bachelor's Thesis (2021 – 2022)
Graz University of Technology (TU Graz)

Point Cloud Reconstruction for Patient-Specific Cranial Implant Design
Harsh Sulakhe · Bachelor's Thesis (2021 – 2022)
BITS Pilani, India · CAML

3D Skull Shape Modelling Using Principal Component Analysis
Lei Yu · Research Project (2020 – 2021)
Graz University of Technology (TU Graz)

06 INVITED LECTURES, SEMINARS & ACADEMIC OUTREACH

2024 Digitale Medizin und Künstliche Intelligenz
Univ. Hospital Essen / UDE · Lecturer: *Medical Shape Analysis*

2024 Host for High-School Research Internships
IKIM, Essen University Hospital · Schülerpraktikum (Gymnasien)

2023 KiMedWS2324 - KI in der Medizin
University of Duisburg-Essen · Teaching Assistant

2023 3D-Druck in Medizin und Zahnmedizin
Uniklinik RWTH Aachen · Lecturer: *AI-Assisted 3D Modeling*

2022 Digitale Medizin und Künstliche Intelligenz
Univ. Hospital Essen / UDE · Lecturer: *Medical 3D Printing*

2020 OpenLabNight 2020 Student Demos
ICG, TU Graz · Research demos & recruitment